Land 13: Gastra Hollows

Digestion and metabolism · OpenStax Ch. 23 and 24

Key ideas

GI tract wall and nervous control

- The alimentary canal runs from mouth to anus; accessory organs (teeth, tongue, salivary glands, liver, gallbladder, pancreas) assist it.
- Wall layers from lumen outward: mucosa (epithelium, lamina propria, muscularis mucosae), submucosa, muscularis, serosa (adventitia in the esophagus).
- The muscularis is usually inner circular and outer longitudinal smooth muscle; the stomach adds an inner oblique layer.
- Enteric nervous system: submucosal plexus mainly regulates secretion; myenteric plexus in the muscularis mainly controls motility.
- Parasympathetic input generally increases digestive activity; sympathetic input decreases it.

Peritoneum and digestive processes

- Visceral peritoneum covers organs; parietal peritoneum lines the abdominal wall; peritoneal fluid reduces friction.
- Mesenteries anchor organs and carry vessels; the greater omentum hangs from the greater curvature, the lesser omentum links stomach and liver.
- Retroperitoneal organs lie behind the peritoneum: most of the duodenum, the pancreas, ascending and descending colon, rectum.
- Six processes: ingestion, propulsion (swallowing and peristalsis), mechanical digestion (chewing, churning, segmentation), chemical digestion, absorption, defecation.
- Peritonitis is inflammation of the peritoneum, often from a ruptured organ such as the appendix.

Mouth, pharynx and esophagus

- Saliva contains salivary amylase (starch), lingual lipase (fat), mucus, lysozyme and IgA; parotid glands are the largest salivary glands.
- Teeth: incisors cut, cuspids tear, premolars and molars crush and grind; enamel covers the crown and is the hardest substance in the body.
- Swallowing has a voluntary oral phase, then involuntary pharyngeal and esophageal phases.
- In the pharyngeal phase the soft palate closes the nasopharynx and the epiglottis covers the larynx.
- The lower esophageal sphincter keeps acid out of the esophagus; when it fails, gastroesophageal reflux disease (GERD) results.

Stomach

- Regions: cardia, fundus, body, pylorus; the pyloric sphincter controls emptying into the duodenum.
- Parietal cells secrete HCl and intrinsic factor; chief cells secrete pepsinogen and gastric lipase; G cells secrete gastrin; mucous cells protect the lining.
- HCl converts pepsinogen to pepsin, which begins protein digestion; acid also kills many microbes.
- Phases of gastric secretion: cephalic (sight, smell, thought), gastric (food stretches the stomach), intestinal (chyme in duodenum, mostly inhibitory).
- Peptic ulcers are most often linked to H. pylori; loss of parietal cells removes intrinsic factor and causes pernicious anemia.

Small and large intestine

- Small intestine: duodenum, jejunum, ileum; most chemical digestion and absorption happen here.
- Surface area is multiplied by circular folds, villi and microvilli (the brush border, which carries brush border enzymes).
- Segmentation mixes chyme against the mucosa; peristalsis propels it; the ileocecal valve opens into the cecum.
- Large intestine: cecum and appendix, ascending, transverse, descending and sigmoid colon, rectum, anal canal.
- The colon absorbs remaining water and electrolytes; its bacteria ferment residue and make vitamin K and some B vitamins.
- Defecation reflex: rectal stretch triggers contraction; the internal anal sphincter is involuntary, the external is voluntary.

Accessory organs and GI hormones

- Liver lobules: blood flows from portal triads through sinusoids to the central vein; bile flows the opposite way in canaliculi.
- Hepatocytes make bile (bile salts, bilirubin, cholesterol); the gallbladder stores and concentrates it; blocked bile flow causes jaundice and pale stools.
- Exocrine pancreas: acinar cells release enzymes (amylase, lipase, nucleases, trypsinogen, chymotrypsinogen); duct cells release bicarbonate.
- Trypsinogen is activated by brush border enteropeptidase; trypsin then activates the other protein digesting enzymes.
- Gastrin stimulates acid; secretin stimulates bicarbonate; CCK stimulates pancreatic enzymes and gallbladder contraction; GIP slows the stomach and boosts insulin release.

Chemical digestion and absorption

- Carbohydrates: amylases make disaccharides; brush border lactase, sucrase and maltase make monosaccharides. Lactase deficiency causes lactose intolerance.
- Glucose and galactose enter by secondary active transport with sodium; fructose by facilitated diffusion; all go to the hepatic portal vein.
- Proteins: pepsin, then trypsin, chymotrypsin and carboxypeptidase, then brush border peptidases yield amino acids.
- Lipids: bile salts emulsify, lipase forms fatty acids and monoglycerides, micelles carry them to the cells, chylomicrons leave through lacteals.
- Fat soluble vitamins travel with micelles; B12 needs intrinsic factor and is absorbed in the ileum; most water is absorbed in the small intestine.
- Celiac disease damages villi and causes broad malabsorption.

Cellular respiration and nutrient metabolism

- Catabolism breaks molecules down and releases energy; anabolism builds molecules and uses energy.
- Glycolysis (cytoplasm) splits glucose into 2 pyruvate with a net 2 ATP and 2 NADH; pyruvate becomes acetyl CoA in mitochondria.
- Krebs cycle (matrix): each acetyl CoA gives 3 NADH, 1 FADH2, 1 ATP and 2 CO2; the electron transport chain uses a hydrogen ion gradient to make most ATP, with oxygen as final acceptor. OpenStax gives a net total of about 36 ATP per glucose; newer estimates are closer to 30 to 32.
- Glycogenesis stores glucose as glycogen; glycogenolysis frees it; gluconeogenesis makes glucose from lactate, glycerol and amino acids.
- Beta oxidation in mitochondria cuts fatty acids into acetyl CoA; excess acetyl CoA in the liver forms ketone bodies.
- Amino acids are deaminated before use for energy; the liver turns the ammonia into urea.

Metabolic states, energy and micronutrients

- Absorptive state: insulin promotes glucose uptake and storage as glycogen, fat and protein.
- Postabsorptive state: glucagon drives liver glycogenolysis and gluconeogenesis; fat becomes the main fuel; prolonged lack of insulin leads to ketoacidosis.
- BMR is energy used at rest for vital functions; carbohydrate and protein supply about 4 kcal per gram, fat about 9.
- The hypothalamus regulates body temperature; heat is lost by radiation, conduction, convection and evaporation.
- Fat soluble vitamins A, D, E, K can be stored; water soluble vitamins (B group and C) are mostly excreted when in excess.
- Major minerals include calcium, phosphorus, sodium, potassium and chloride; trace minerals include iron, iodine and zinc.

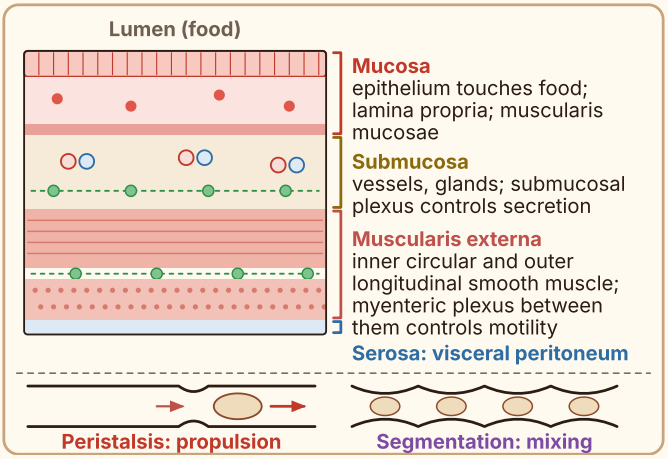
Villager wisdom

- Down in Gastra Hollows we say the gut wall is four blankets thick: mucosa next to the meal, submucosa with the pipes, muscularis that squeezes, and serosa on the outside.
- The stomach keeps three kinds of workers, traveler. Parietal folk pour the acid and intrinsic factor, chief folk bring pepsinogen, and G folk shout gastrin to wake them all.

- Remember the duodenum's two messengers: secretin sends soda (bicarbonate) to cool the acid, and CCK calls the gallbladder to squeeze out bile for the fat.
- Fat takes the back road home. Sugars and amino acids ride the portal vein to the liver, but chylomicrons slip into the lacteals and travel by lymph.
- When the larder is full, insulin stores the harvest; when it runs low, glucagon opens the liver's granary. Burn fat too long with no insulin and ketones sour the blood.

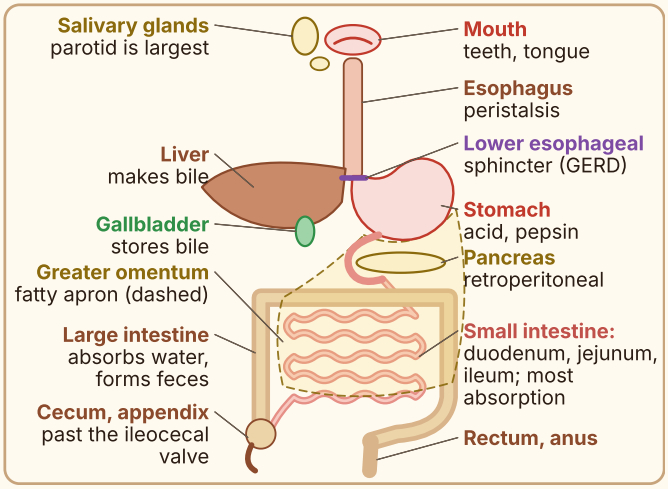
Diagrams

Layers of the GI tract wall



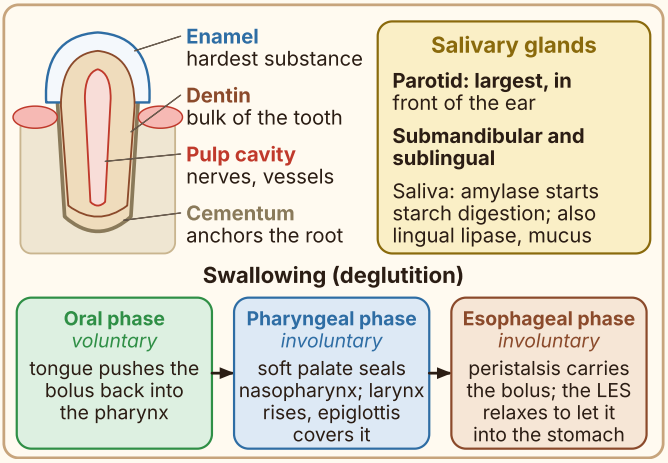
From the lumen outward the wall is mucosa, submucosa, muscularis externa and serosa; the submucosal plexus mainly controls secretion and the myenteric plexus, between the muscle layers, controls motility.

The digestive system



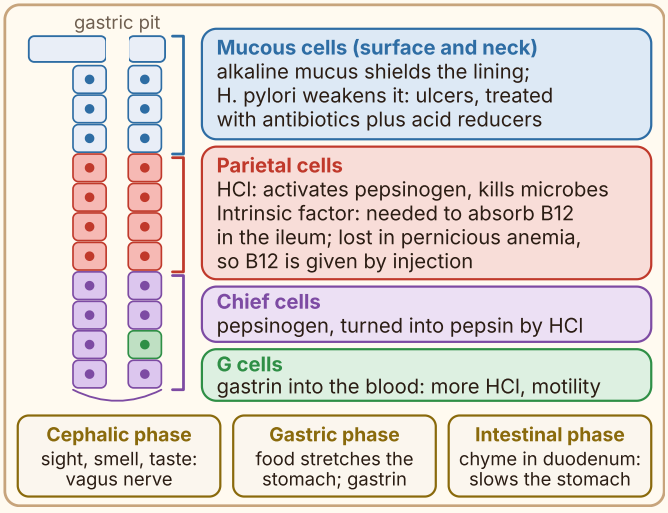
Food passes from the mouth through the pharynx, esophagus, stomach, small intestine and large intestine; the salivary glands, liver, gallbladder and pancreas add secretions. Most of the 9 L of fluid entering the gut each day is absorbed in the small intestine.

Mouth, teeth and swallowing



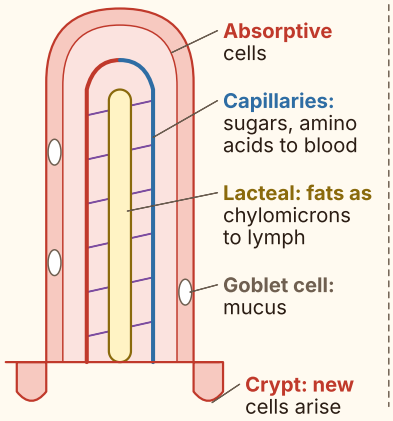
Enamel is the hardest substance in the body and saliva starts starch digestion with salivary amylase. Only the oral phase of swallowing is voluntary; in the pharyngeal phase the larynx rises and the epiglottis covers it.

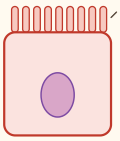
Gastric glands and their secretions



Parietal cells secrete HCl and intrinsic factor, chief cells secrete pepsinogen that acid turns into pepsin, and G cells release gastrin into the blood. The intestinal phase slows the stomach once chyme reaches the duodenum.

Villus and brush border





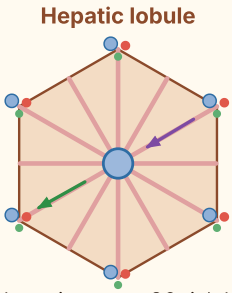
Brush border enzymes: lactase, sucrase, maltase, peptidases and enteropeptidase

Low lactase: lactose intolerance

Celiac disease: gluten flattens villi, so less is absorbed

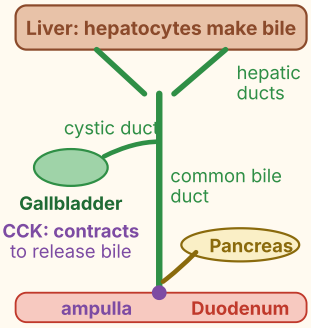
Circular folds, villi and microvilli give the small intestine a huge absorptive surface; sugars and amino acids enter the capillaries while fats enter the lacteal. Flattened villi in celiac disease cut that surface and cause malabsorption.

Liver lobule and bile flow



- hepatic artery: O2 rich blood
- portal vein: blood from the gut
- bile duct
- central vein: to hepatic veins

Purple: blood to the central vein
Green: bile out to the triads



Bile salts emulsify fats
Stone in the common bile duct: jaundice (bilirubin) and pale, fatty stools

Blood from the hepatic artery and hepatic portal vein flows through the sinusoids to the central vein, while bile flows the opposite way to the bile ducts. CCK makes the gallbladder contract; a stone in the common bile duct causes jaundice and pale, fatty stools.

Digestive hormones

Hormone	Source and trigger	Main effects
Gastrin	G cells, stomach; food, stretch, vagus nerve	more HCl and gastric motility
Secretin	duodenum; acidic chyme	bicarbonate rich pancreatic juice; slows the stomach
CCK	duodenum; fats and proteins in chyme	gallbladder contracts; enzyme rich pancreatic juice; slows emptying
GIP	duodenum; glucose and fat in chyme	boosts insulin release, so oral glucose raises insulin more than IV

Gastrin from the stomach boosts acid secretion, while secretin, CCK and GIP from the duodenum answer acid, fat and glucose in chyme. GIP explains why glucose by mouth releases more insulin than the same glucose by vein.

Absorptive and postabsorptive states

Absorptive (fed)
insulin dominates

Glucose uptake
cells take in glucose for ATP

Glycogenesis
glucose stored as glycogen: liver, muscle

Lipogenesis
extra glucose and fat stored in adipose

Protein synthesis
amino acids build proteins

Anabolism: builds, uses energy

Postabsorptive (fasting)
glucagon dominates

1 Glycogenolysis (first)
liver glycogen to glucose

2 Gluconeogenesis
new glucose from lactate, glycerol, amino acids

3 Lipolysis
fatty acids become the main fuel, sparing glucose

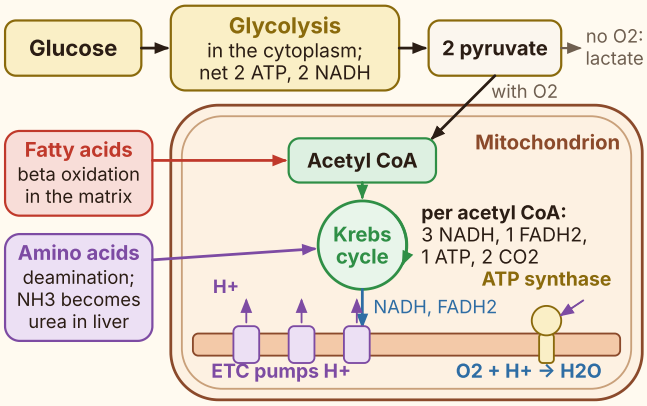
4 Ketogenesis
extra acetyl CoA in liver becomes ketone bodies

Catabolism: releases energy

Untreated type 1 diabetes: no insulin, so cells burn fat;
ketones pile up and pH falls (ketoacidosis, fruity breath)

After a meal insulin stores fuel as glycogen, fat and protein; between meals glucagon keeps blood glucose up, first by glycogenolysis and then by gluconeogenesis, while fat becomes the main fuel. Without insulin, as in untreated type 1 diabetes, fat burning floods the blood with ketones.

Cellular respiration and fuel entry



Oxygen is the final electron acceptor, forming water.

Total: about 30 to 36 ATP per glucose,
over 90 percent of it made in the mitochondria

Glycolysis in the cytoplasm nets 2 ATP; the Krebs cycle and the electron transport chain in the mitochondria make the rest, with oxygen as the final electron acceptor. Fatty acids enter as acetyl CoA after beta oxidation, and amino acids enter after deamination.

Vitamins and minerals

Fat soluble: A, D, E, K

- **A** vision, epithelia
- **D** calcium absorption
- **E** antioxidant
- **K** clotting factors

absorbed with fat in micelles; stored in liver and fat, so excess can build up

Bile blocked: low K, bruising

Water soluble: B, C

- **B** metabolic coenzymes
- **B12** needs intrinsic factor; absorbed in the ileum
- **C** collagen, antioxidant

little is stored (except B12); excess is lost in urine, so they are needed often

Minerals

Calcium: bones, muscle, clotting; absorbed with vitamin D
Iron: hemoglobin. Iodine: thyroid hormone.
Sodium, potassium: nerve and muscle signals
Low vitamin D: little Ca2+ absorbed, weak bones

Fat soluble vitamins are absorbed with dietary fat and stored in the liver and fat tissue, so poor fat absorption causes deficiencies such as low vitamin K and easy bruising. Water soluble vitamins are stored little and excess leaves in urine; B12 needs intrinsic factor to be absorbed in the ileum.

Food energy and heat balance

Energy per gram of food

Carbohydrate		4 kcal per g
Protein		4 kcal per g
Fat		9 kcal per g

50 g carb × 4 + 20 g protein × 4 + 10 g fat × 9
= 200 + 80 + 90 = about 370 kcal

Basal metabolic rate: energy used at rest to keep vital functions going; raised by thyroid hormone and more muscle

Heat loss (hypothalamus is the thermostat)

Radiation	Conduction	Convection	Evaporation
infrared to cooler objects	direct contact, as a cold chair	moving air or water	sweat; slowed by humid air

Carbohydrate and protein give 4 kcal per gram and fat gives 9, so 50 g carbohydrate, 20 g protein and 10 g fat give about 370 kcal. The body sheds heat by radiation, conduction, convection and evaporation, and humid air slows evaporation of sweat.

Chemical digestion and absorption

	Enzymes and where they act	Products	Absorbed
Carbs	Mouth: salivary amylase Pancreas: pancreatic amylase Brush border: lactase, sucrase, maltase	glucose, galactose, fructose	into blood; glucose with Na+
Proteins	Stomach: pepsin Pancreas: trypsin, chymotrypsin (trypsinogen is activated by enteropeptidase) Brush border: peptidases	amino acids, small peptides	with Na+ into blood

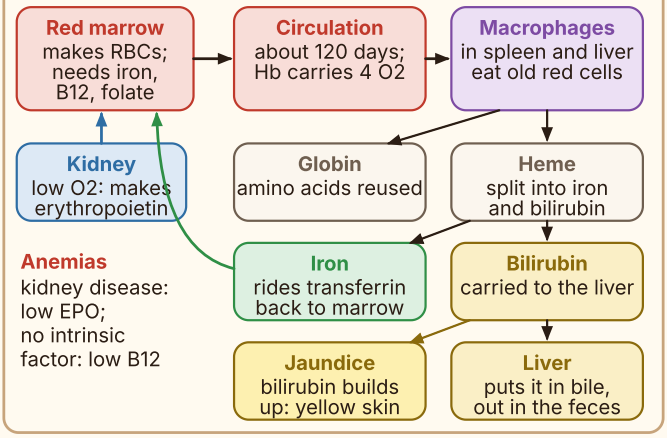
Fats



fat droplet → bile salts emulsify → micelles carry fatty acids → cell makes chylomicrons → lacteal (lymph)

Carbohydrates end as monosaccharides and proteins as amino acids, both absorbed into blood capillaries, glucose and amino acids together with Na+. Fats are emulsified by bile, split by lipase, carried in micelles and leave the cell as chylomicrons that enter the lacteals.

Life cycle of a red blood cell



Red cells circulate about 120 days; macrophages recycle them, iron returns to the marrow on transferrin, and bilirubin goes to the liver and out in bile, so a buildup of bilirubin causes jaundice.

Self check

1. **After drinking milk, a patient gets bloating, gas and diarrhea. Deficiency of which enzyme, located where, is most likely?**
    A. Amylase in pancreatic juice
    B. Pepsin in gastric juice
    C. Lipase in pancreatic juice
    D. Lactase on the brush border
2. **Which cells in the stomach secrete the hormone gastrin into the blood?**
    A. Chief cells
    B. G cells
    C. Parietal cells
    D. Mucous neck cells
3. **Glucose taken by mouth triggers more insulin release than the same amount given by vein. Which duodenal hormone helps explain this?**
    A. Gastrin
    B. Secretin
    C. GIP
    D. Glucagon
4. **A person with untreated type 1 diabetes has high blood glucose, fruity breath and low blood pH. The best explanation is:**
    A. Excess insulin drives glucose into cells
    B. Muscle glycogen breakdown releases acid
    C. Without insulin, fat is burned and ketones rise
    D. Excess deamination makes too much urea
5. **A few hours after a meal, absorption is complete. What first keeps blood glucose steady?**
    A. Muscle glycogen released into blood
    B. Conversion of ketones into glucose
    C. Insulin driven glycogenesis
    D. Liver glycogenolysis driven by glucagon
6. **Most ATP made from glucose is formed by ATP synthase, powered by:**
    A. Substrate level phosphorylation in glycolysis
    B. A hydrogen ion gradient across the inner membrane
    C. Direct transfer of phosphate from FADH2
    D. Glycogen breakdown in the cytoplasm
7. **Body heat moving from your skin directly into a cool metal chair you sit on is heat loss by:**
    A. Conduction
    B. Convection
    C. Radiation
    D. Evaporation

8. **Glucagon raises blood glucose mainly by stimulating the liver to:**
    A. Take up glucose and build glycogen
    B. Release glucose stored in muscle
    C. Convert glucose into triglycerides
    D. Break down glycogen and make new glucose
9. **A gallstone completely blocks a patient's common bile duct. Which pair of findings is most expected?**
    A. Bloating and diarrhea after milk
    B. Low vitamin B12 and anemia
    C. Jaundice and pale, fatty stools
    D. Excess gastric acid and heartburn
10. **In the absorptive (fed) state, which hormone dominates, and what does it do?**
    A. Glucagon, promoting glycogen breakdown
    B. Insulin, promoting gluconeogenesis
    C. Insulin, promoting glucose uptake and storage
    D. Cortisol, promoting protein breakdown
11. **Pepsinogen secreted by chief cells is converted to active pepsin mainly by:**
    A. Enteropeptidase on the brush border
    B. Intrinsic factor
    C. Hydrochloric acid in the stomach
    D. Bile salts
12. **In the electron transport chain, oxygen serves as the:**
    A. First electron donor
    B. Carbon source for released CO2
    C. Final electron acceptor, forming water
    D. Enzyme that splits glucose

Answer key

1. **D.** Lactase on the brush border. Lactase is a brush border enzyme; without it lactose stays in the lumen, draws water osmotically and is fermented by colon bacteria, causing gas. Pancreatic amylase digests starch, not the disaccharide lactose.
2. **B.** G cells. G cells, found mainly in the pyloric antrum, release gastrin, which stimulates acid secretion and gastric motility. Parietal cells respond to gastrin by making HCl but do not secrete it.
3. **C.** GIP. GIP is released by the duodenum when glucose and fat arrive and stimulates the pancreas to release insulin, so oral glucose produces a larger insulin response. Secretin is also duodenal, but its main action is bicarbonate secretion, not insulin release.
4. **C.** Without insulin, fat is burned and ketones rise. Without insulin, cells cannot take up glucose, so fat is broken down and the liver makes ketone bodies, which are acids that lower blood pH. Excess insulin would cause low, not high, blood glucose.
5. **D.** Liver glycogenolysis driven by glucagon. Early in the postabsorptive state, glucagon causes the liver to break down glycogen and release glucose. Muscle glycogen is tempting, but muscle lacks glucose 6 phosphatase and keeps its glucose for its own use.

6. **B.** A hydrogen ion gradient across the inner membrane. The electron transport chain pumps hydrogen ions into the intermembrane space, and their flow back through ATP synthase drives oxidative phosphorylation. Substrate level phosphorylation in glycolysis and the Krebs cycle makes only a small share of the ATP.
7. **A.** Conduction. Conduction transfers heat between objects in direct contact. Convection is tempting, but it requires heat to be carried away by moving air or water, not a solid surface.
8. **D.** Break down glycogen and make new glucose. Glucagon triggers glycogenolysis and gluconeogenesis in liver cells, which release glucose into the blood. Taking up glucose and building glycogen is the action of insulin, the opposing hormone.
9. **C.** Jaundice and pale, fatty stools. Blocked bile cannot reach the duodenum, so bilirubin backs up into blood (jaundice), stools lose their brown pigment, and fat is poorly digested. Bloating after milk points to lactase deficiency, a brush border problem unrelated to bile flow.
10. **C.** Insulin, promoting glucose uptake and storage. After a meal, rising blood glucose triggers insulin, which helps cells take up glucose and stimulates glycogen, fat and protein synthesis. Insulin inhibits rather than promotes gluconeogenesis, so pairing insulin with gluconeogenesis names the right hormone but the wrong action.
11. **C.** Hydrochloric acid in the stomach. The low pH from HCl cleaves pepsinogen into pepsin, and pepsin then activates more pepsinogen. Enteropeptidase is an activating enzyme too, but it acts in the small intestine on trypsinogen, not in the stomach.
12. **C.** Final electron acceptor, forming water. Oxygen accepts electrons at the end of the chain and combines with hydrogen ions to form water. The carbon in exhaled CO_2 comes from glucose carbons in the Krebs cycle and pyruvate oxidation, not from oxygen gas.

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